Group Assignment 1: Data Preprocessing Report

Statement of Contribution

Deliverable	Sebastian	Gary	Lenard
Program Design	All	All	All
Data Import	All	All	All
Data Preparation	All	All	All
Data Preprocessing	All	All	All
Final visualization	All	All	All
Debugging	All	All	All
Part B	All	All	All
Report	All	All	All

All group members met together to brainstorm preprocessing techniques and compare figures and plotting, program design, batch processing, debugging, etc.

By submitting this assignment all group members agree to the above listed contributions to be true and accurate.

PART A

Program Design:

1. Data preparation
    a. Load donor.tsv into MATLAB and import icgc_donor_id, donor_diagnosis_icd10
    b. Load mirna_seq.tsv into MATLAB and import icgc_donor_id, icgc_sampled_id, mirna_id, normalized_read_count, raw_read_count
    c. Identify DLBCL samples with ICD10 code C83.3 and filter donor table for C83.3 diagnosis
2. Data extraction and formatting
    a. Extract DLBCL donor IDs from filtered donor table
    b. Filter mirna_seq table to keep only DLBCL samples with innerjoin to match donor IDs
    c. Reformat raw count data using reformatICGCdata() to convert miRNA (rows) x samples (columns)
    d. Reformat normalized count data with reformatICGCdata()

    e. Save workspace so variables are ready for next iteration
3. Initial visualization for raw and normalized counts
    a. Grab expression matrices from the cell arrays
    b. For each sample, we calculate the QC metrics:
        i. Total counts per sample
        ii. IQR per sample
        iii. Expression distribution per sample with boxplots
    c. Plots for visualization
        i. Total counts scatter plot
        ii. IQR scatter plot
        iii. Boxplots (log2 transformed to visualize after we do the initial visualization)
    d. Compare raw vs normalized data figures
4. Normalization (for raw counts)
    a. Apply Relative Frequency normalization with SampleNormalizationRF()
    b. Replace zero values before log transformation with replaceZeros(data, ‘lowval’)
    c. Apply log2 transformation where log2_data = log2(data_with_replaced_zeros)
5. Post-normalization visualization
    a. Look at skew distribution to see if transformation became more normal
    b. Generate boxplots for RF normalized data and the provided normalized data
    c. Compare normalization methods through visualization through better box alignment, reasonable IQR, etc. to choose best one
6. Quality Control on chosen normalized data
    a. Calculate QC metrics for IQR per sample and average Spearman correlation with SampleCorrelation()
    b. Find outliers with scatterplotMarkOutliers()
        i. Calculate alpha with computeAlphaOutliers()
        ii. Plot IQR with outlier threshold
        iii. Plot average correlation with outlier threshold
    c. Check for batch effect patterns in processing order
    d. Identify if any samples need to be removed, such as IQR = 0, technical issues, or if below correlation threshold by more than 0.2 (removal through boolean mask)
    e. Remove outlier samples if needed with a boolean mask where samples greater than 0.2 under the alpha threshold are removed
    f. Repeat QC in general until outliers are not present
7. Filter low-expressed miRNAs
    a. Find a quantile threshold for filtering (Q = 0.90, etc.)
    b. Apply MarkLowCounts() to keep only miRNAs expressed above threshold in at least one sample or more
    c. Filter the expression matrix to remove low-expressed features
8. Final visualization and QC
    a. Generate boxplots with filtered data
    b. Calculate final QC results and QC summary plots:

i. IQR per sample
ii. Total expression per sample
iii. Median expression distribution per sample

c. Final clustergram with heatmap for more visualization

9. Export processed data

REPORT

Data from patients with Diffuse Large B-Cell Lymphoma were extracted from donor and miRNA sequencing datasets based on the ICD-10 code C83.3 (ICD-10, n,d,). Initial visualization was created for both raw counts and normalized counts based on miRNA sequencing data for each DLBCL sample. Total read counts, IQR, and median was calculated for each sample. Figure 1 -3 illustrate the total read counts, IQR, and a boxplot for the raw data for each sample. Figure 3 -5 illustrate the total read counts, IQR, and a boxplot for the normalized data for each sample.

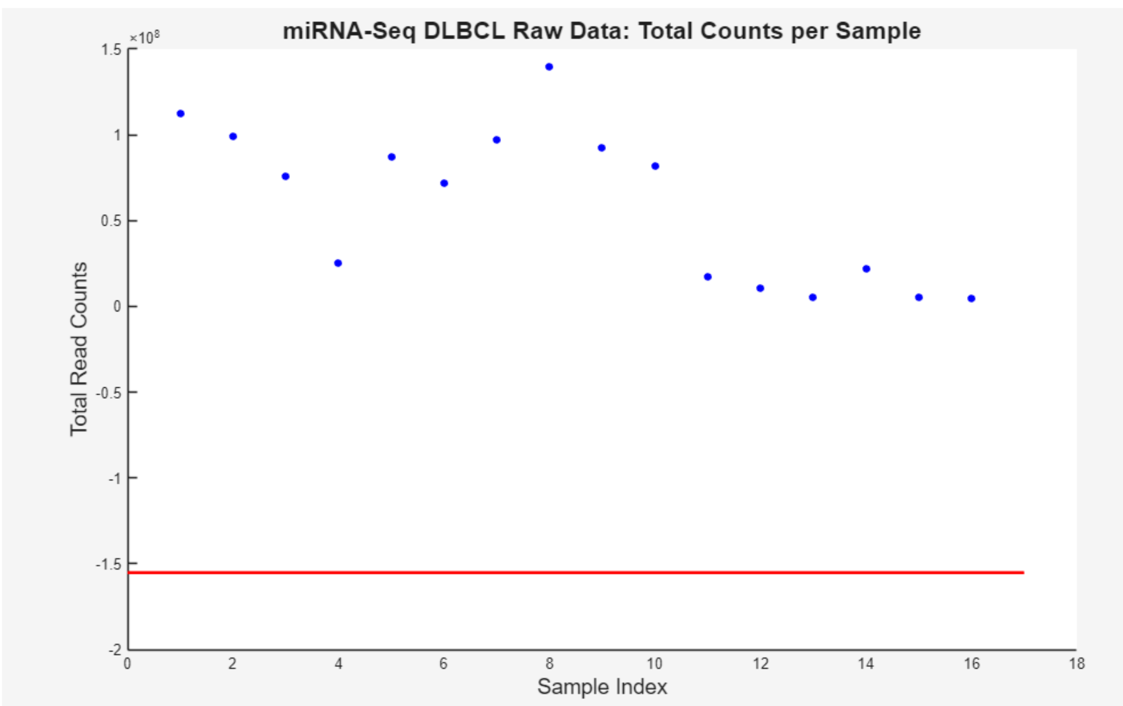

Figure 1. Scatterplot of raw miRNA sequencing data, illustrating total read counts of miRNA for each sample.

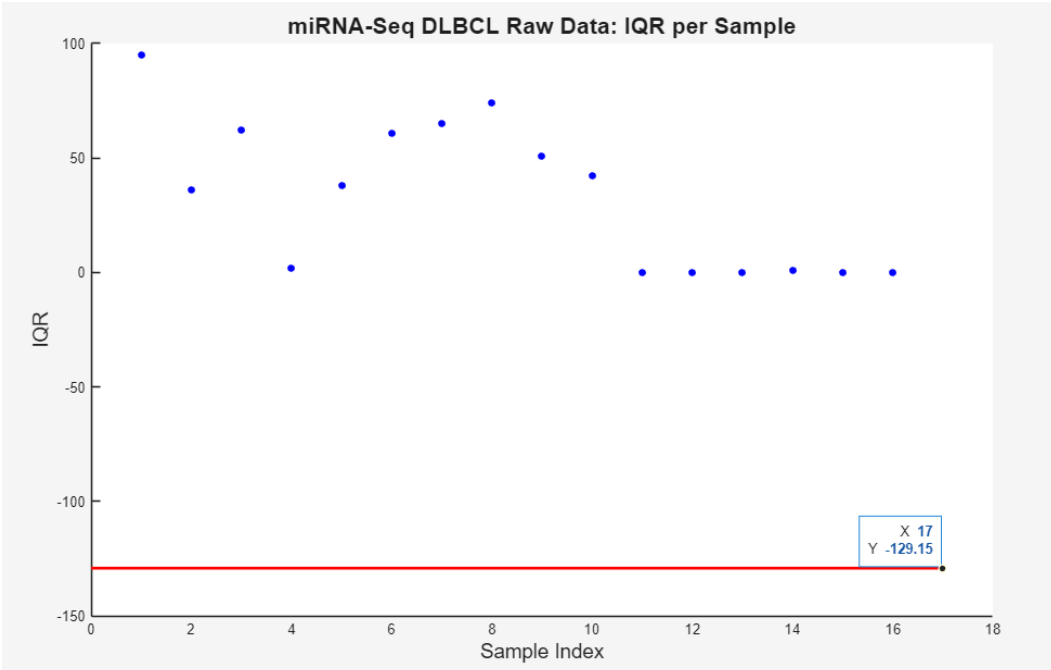

Figure 2. Scatterplot of raw miRNA sequencing data, illustrating IQR of miRNA for each sample.

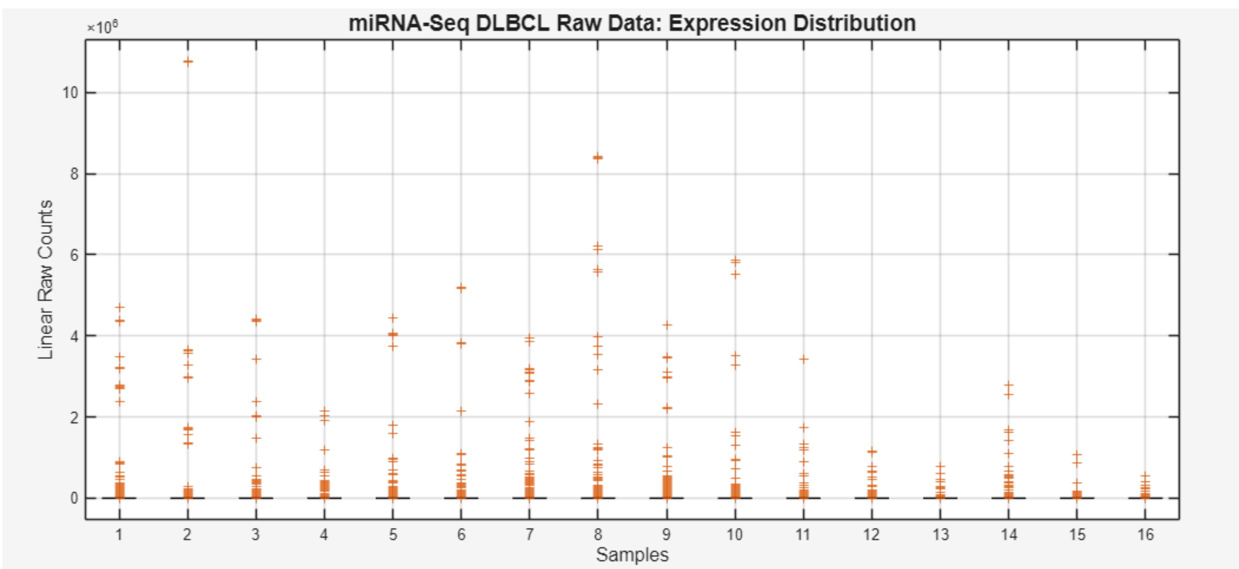

Figure 3. Boxplots of raw miRNA sequencing data, illustrating miRNA expression distribution counts for each sample.

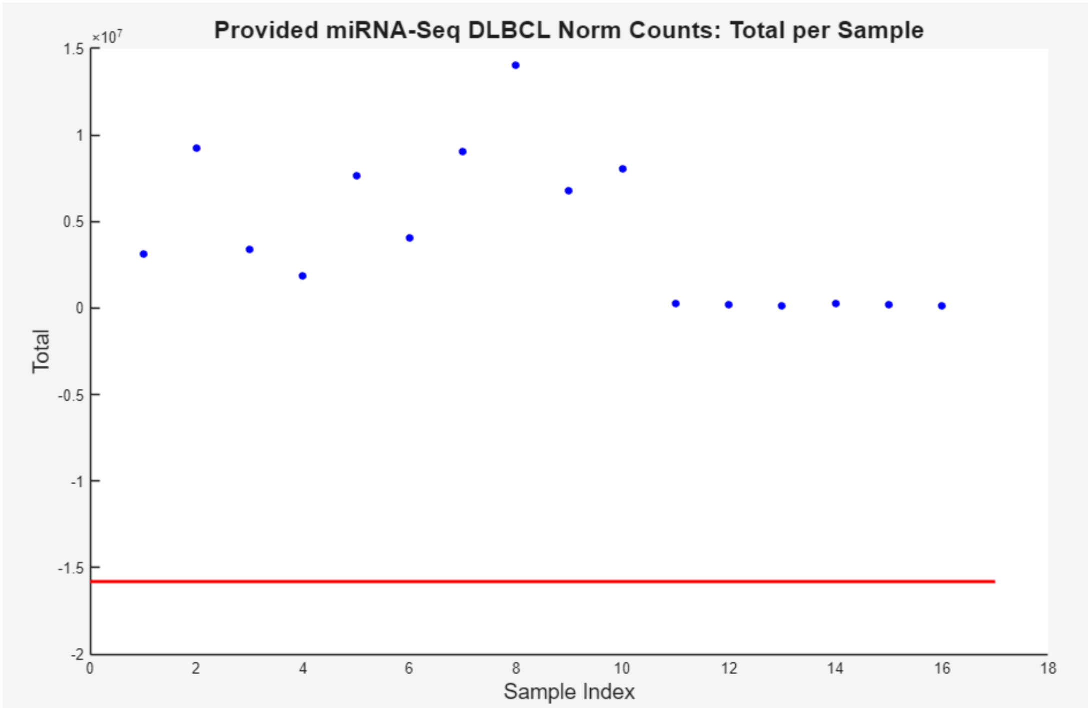

Figure 4. Scatterplot of normalized miRNA sequencing data, illustrating total read counts of miRNA for each sample.

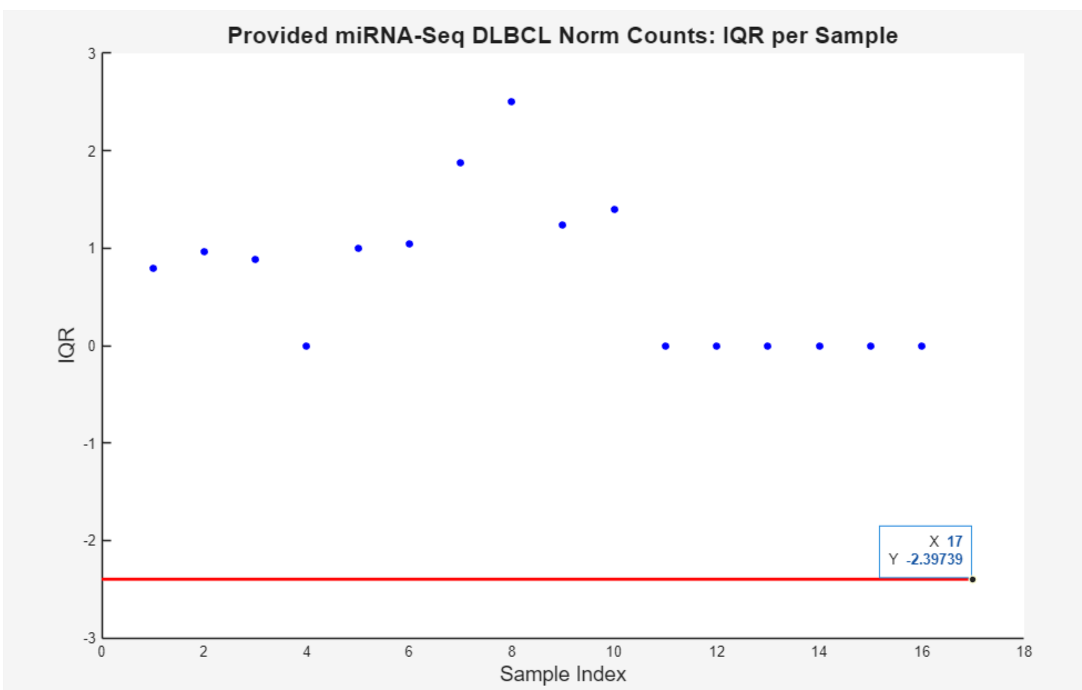

Figure 5. Scatterplot of normalized miRNA sequencing data, illustrating IQR of miRNA for each sample.

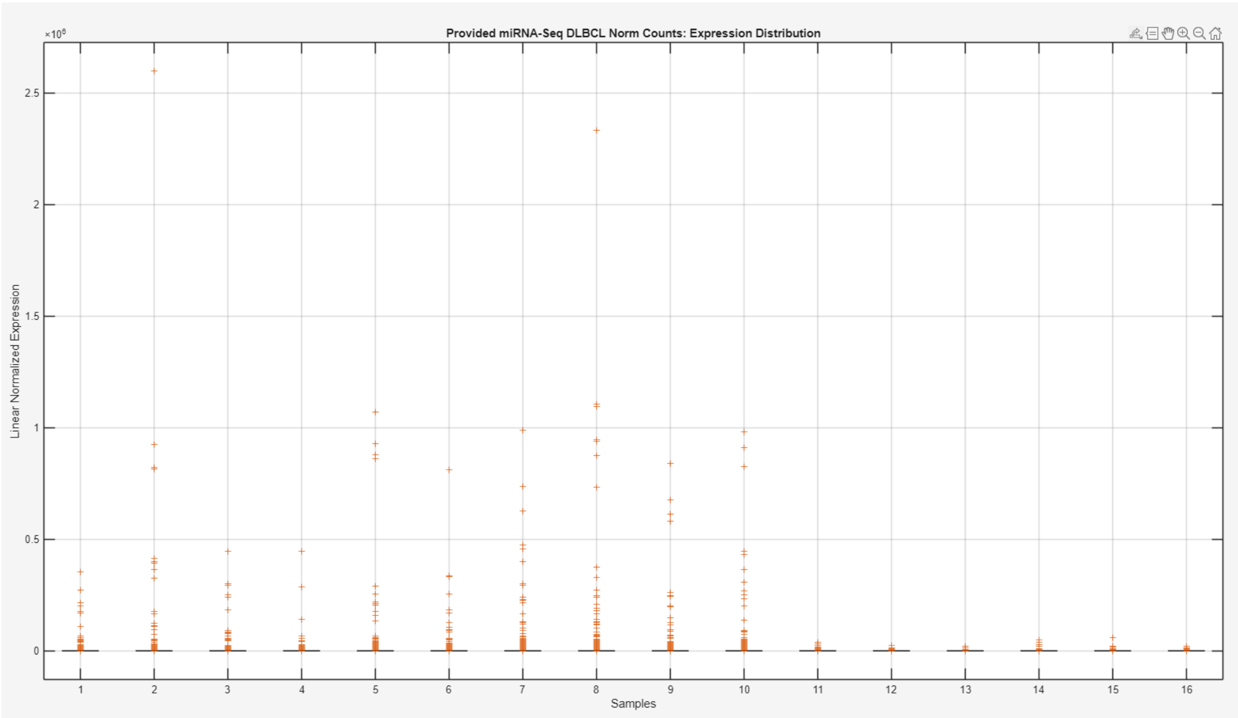

Figure 6. Boxplots of normalized miRNA sequencing data, illustrating miRNA expression distribution counts for each sample.

Most prominently seen in Figure 2 and Figure 5 illustrating the IQR of DLBCL samples, samples 11 to 16 demonstrate little to no variation, with an IQR of approximately 0.

When further investigating Figure 3 and Figure 6, depicting a graph of boxplots for the raw and normalized data respectively, the graph hints that the data is skewed with no appearance of boxes and whiskers for each sample. To investigate this idea further, a histogram was used to explore the skewness of the data. This is graphed in Figure 7 and Figure 8.

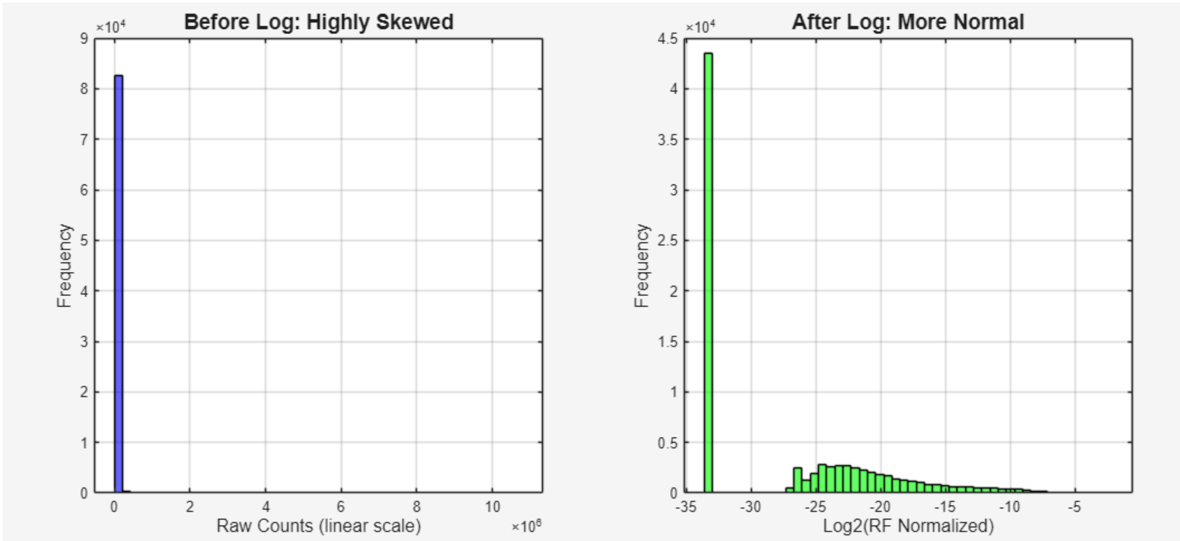

Figure 7. Histograms of raw miRNA total read counts.

As seen in the raw distribution graph in figure 7, the raw miRNA read counts show a highly right-skewed distribution, with the vast majority of values concentrated around zero, with a smaller number having counts above 10^5. This shows before normalization, there are samples that have sequencing depth that dominates the dynamic range, and makes direct comparisons unreliable.

After Relative Frequency normalization with a $\log_2$ transformation, the distribution is more symmetrical and compact. There is less variance, and homogeneity is improved across expression levels. The transformation compresses high outliers while also expanding low-expression regions, this results in a more normalized spread for correlation, clustering, and outlier detection.

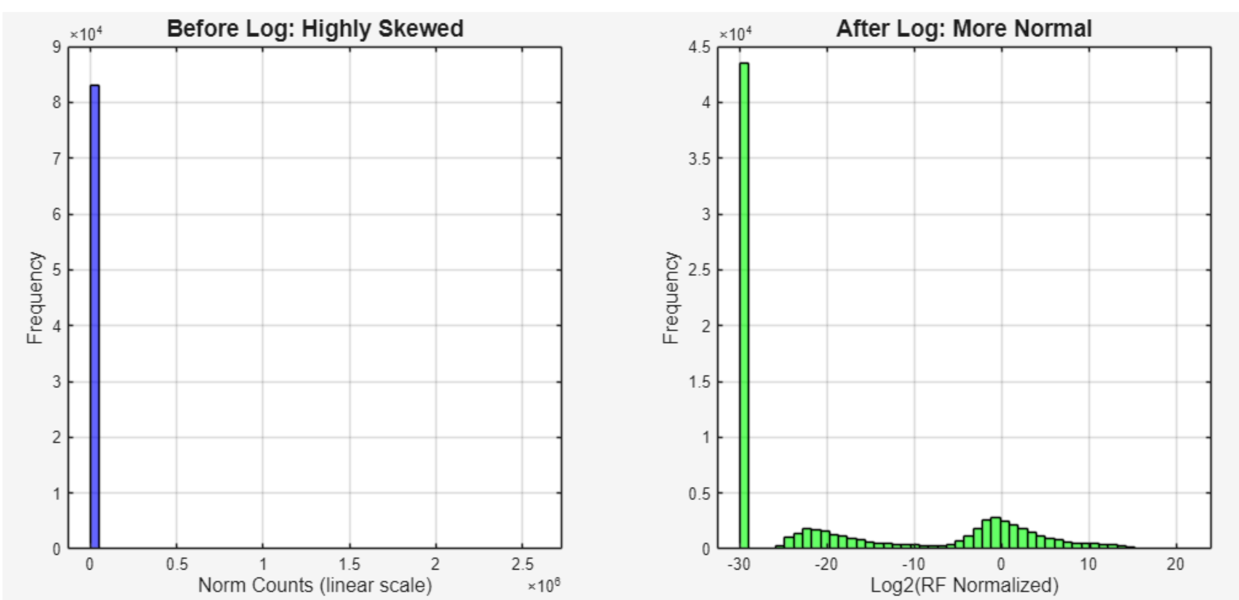

Figure 8. Histogram of before and after transformation of normalized miRNA total read counts.

As seen in Figure 8, a similar distribution is demonstrated before transformation in the normalized miRNA read counts, with a right-skewed distribution where values are near zero. After transformation with log_2, two distributions of lower expressed values are found, while outliers to the dataset are found distanced away at -30.

Normalization of miRNA raw data was performed using the function SampleNormalizationRF. Both raw data that was normalized using the function described, and normalized miRNA data from the ICGC database were compared. This is demonstrated in Figure 9, where a box plot for each normalization method was created to display the various samples.

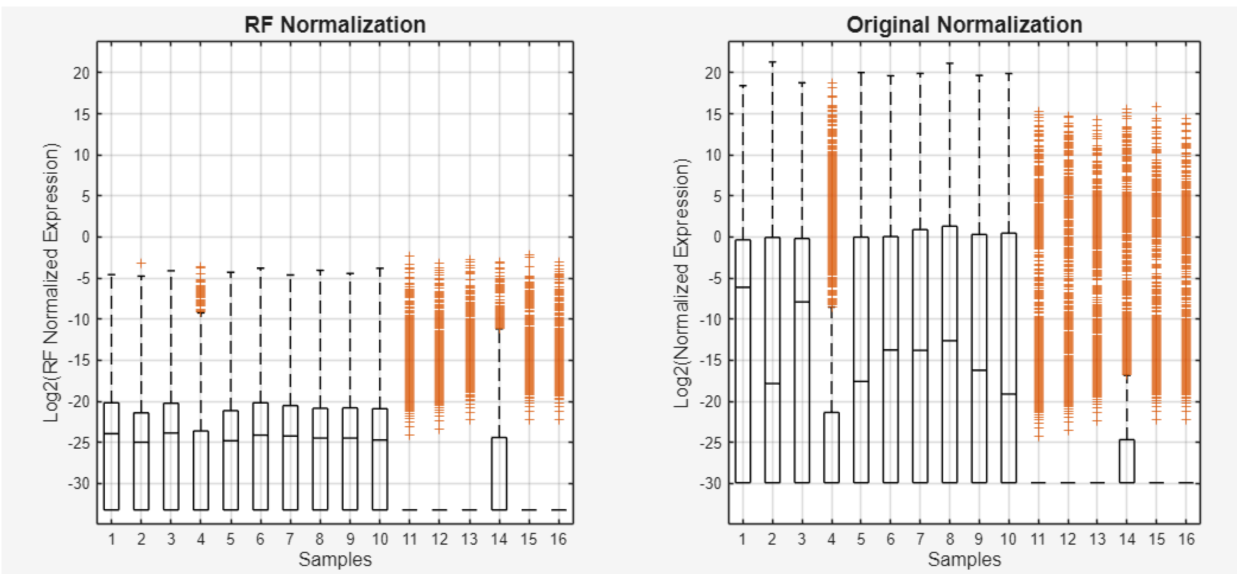

Figure 9. Comparison of normalization using SampleNormalizationRF function and normalization method of ICGC database.

As seen in Figure 9, using the relative frequency normalization function illustrates better alignment of the boxes (IQR) in comparison to the ICGC method. Moreover, there is better alignment of the median values when using the relative frequency normalization function in comparison to the ICGC method. As such, SampleNormalizationRF was determined to be a better normalization method, and miRNA data normalized using this function was chosen to undergo further pre-processing.

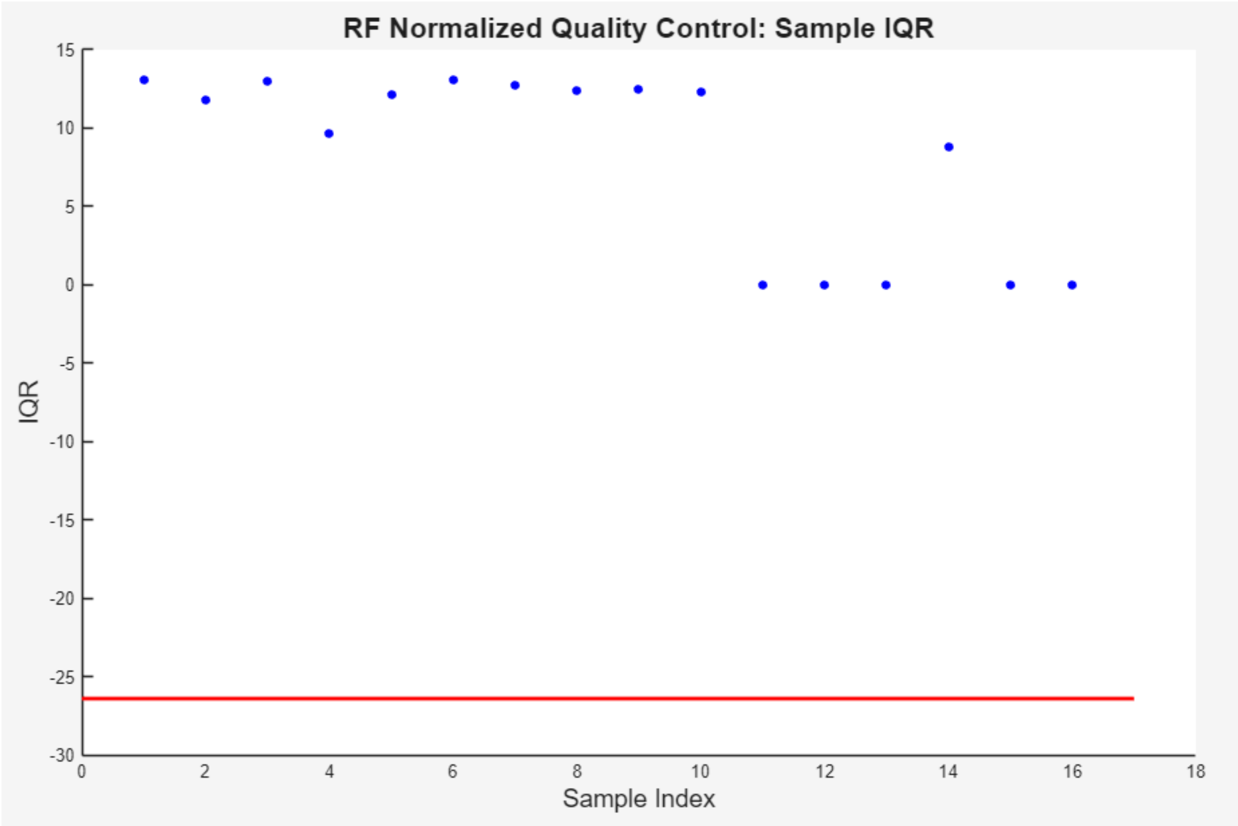

Figure 10. Interquartile range of RF-normalized miRNA samples

As seen in figure 10, the RF-normalized dataset displays two distinct samples while examining their IQR values. It can be seen that samples 1-10 and 14 show a consistent IQR between approximately 9 and 15, which shows that normal biology varies in expression levels across miRNA. In contrast, samples from 11 to 13, and samples 15 to 16 show an IQR of 0, indicating that there is no detectable variation in samples. IQR's of zero implies that all the values are identical or nearly identical, and should be labeled as outliers and removed. This may illustrate potential batch effects with an issue arising later in the data collection.

The red threshold line in figure 10 is the computed lower bound for outlier detection; all samples below this threshold, especially samples with IQR of zero, are to be flagged as technical outliers and to be excluded from further analysis. The removal of low variance samples ensures that the remaining data accurately reflects meaningful biological expressions.

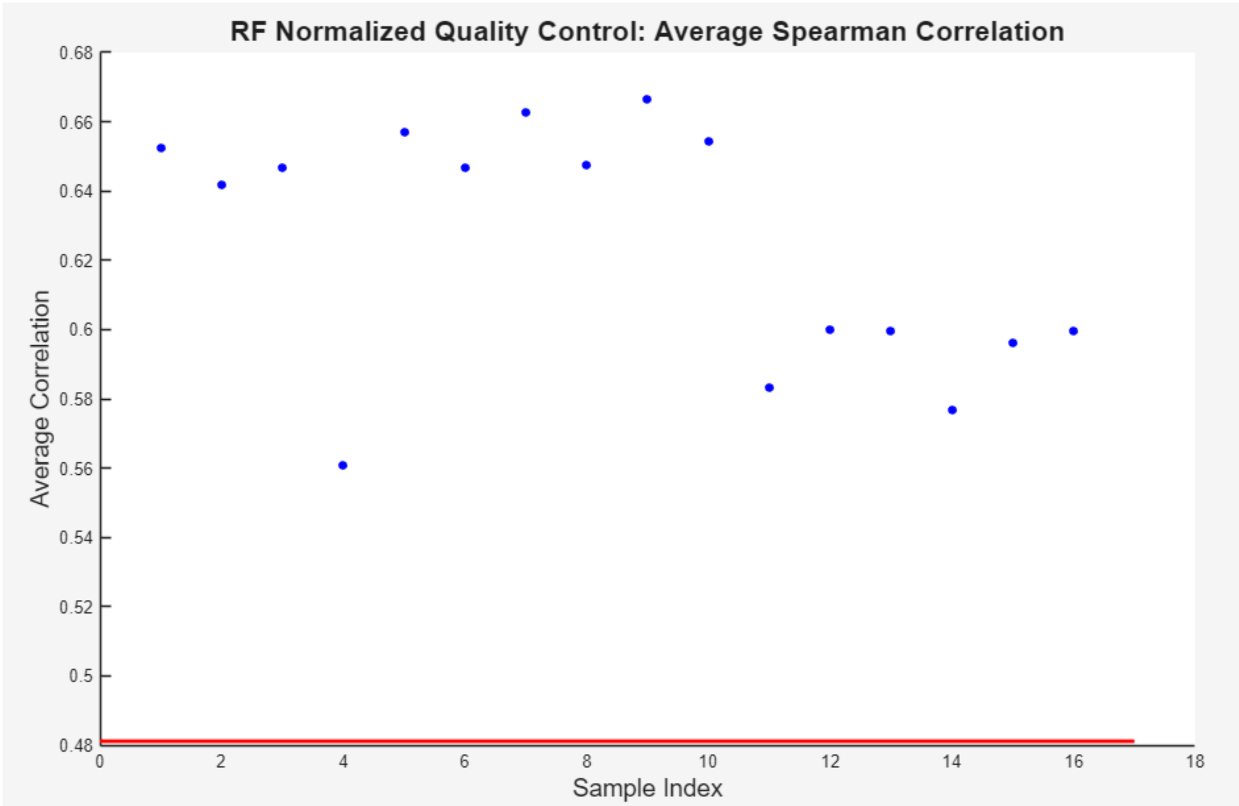

Figure 11. Average Spearman correlation between miRNA samples

Seen in Figure 11, the average Spearman correlation across the RF-normalized datasets varies between 0.56 and 0.67 indicating strong agreement for expression profiles between samples. Samples that have higher average correlations (between 0.64 and 0.67) have consistent miRNA expression relative to other samples, which suggest successful normalization.

However, samples at indexes 11 to 16 have lower correlation values, deviating from the main cluster. These samples fall closer to the lower threshold that is indicated by the red line(0.48). Again, this can illustrate potential batch effects with these samples creating its own cluster. Correlation values near or below the lower threshold can signify potential technical outliers. Figure 12 further illustrates that the IQR of samples 11 to 13, and samples 15 and 16 is 0.

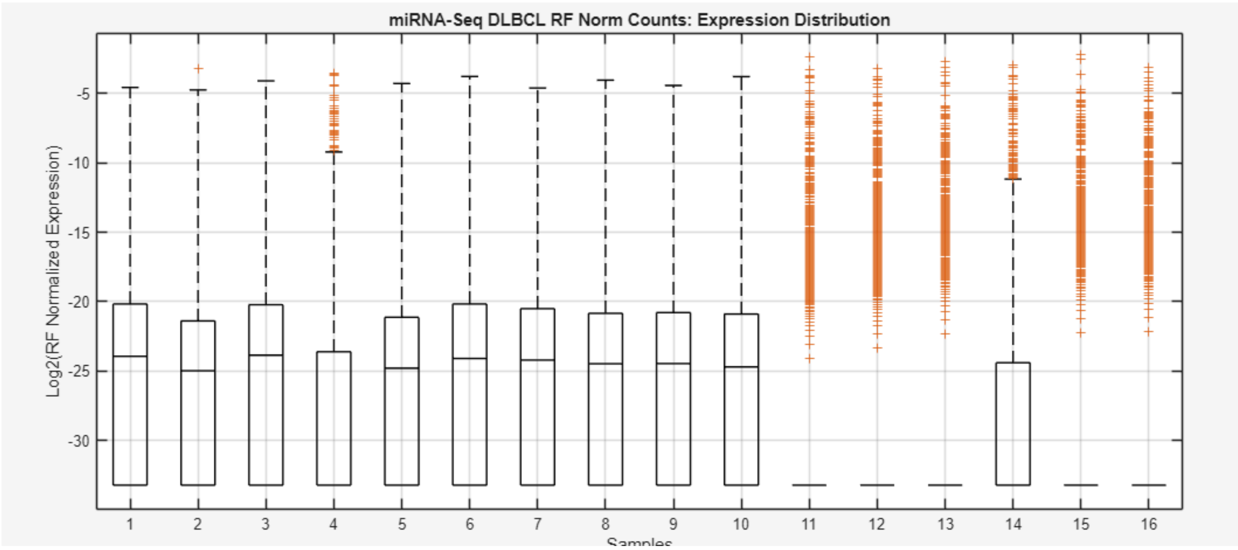

Figure 12. Boxplot of RF normalized miRNA samples.

Combining Figure 10, 11, and 12 analysis, the correlation plot, the IQR graph and the boxplot provides complementary evidence for identifying low quality or abnormal samples due to technical variability. Samples that exhibit IQR = 0 were removed to ensure following clustering, comparisons, and statistical testing are based on meaningful variation rather than noise.

After removal of samples with IQR = 0, visualization was performed again as seen in Figure 13, 14, and 15.

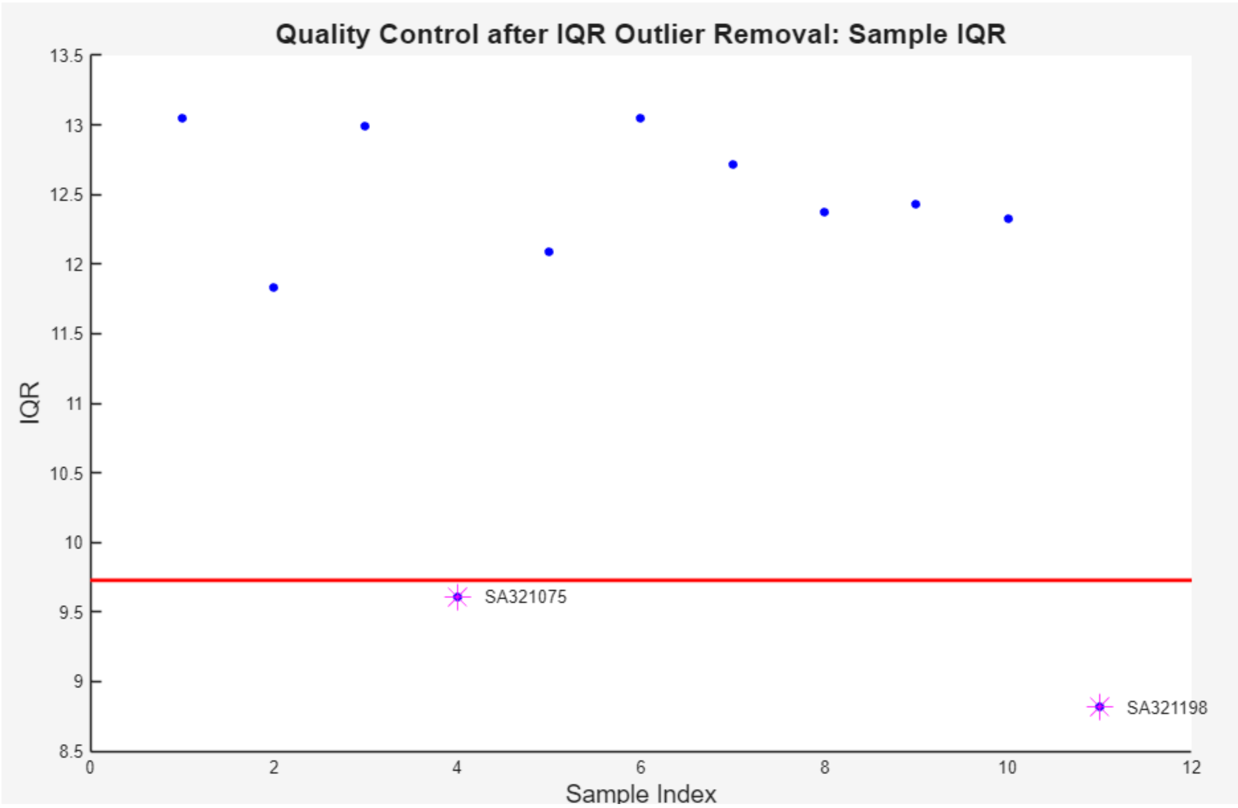

Figure 13. Scatterplot of IQR of samples after removal of samples with IQR = 0.

Most evident in Figure 13, following the removal of the IQR = 0 samples from the previous step, the interquartile range values stabilize for the remaining samples between 9.5 – 13.5, reflecting better variance consistency across the data. Two samples (SA321075 and SA321198) lie just beneath the computed threshold (red line), making them low-variability candidates that should receive further scrutiny. These observations informed the subsequent filtering step, which used both IQR and correlation as checks to confirm true outliers.

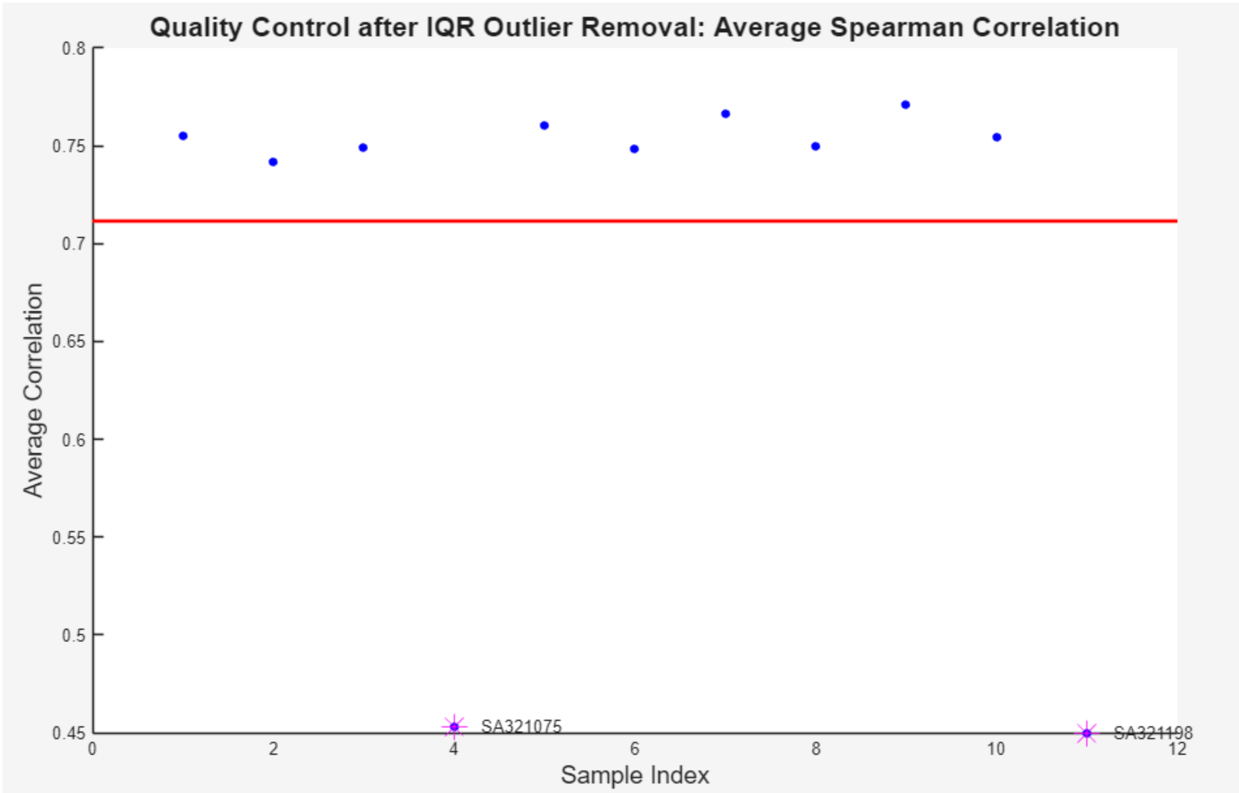

Figure 14. Scatterplot of average correlation of samples after removal of samples with IQR = 0.

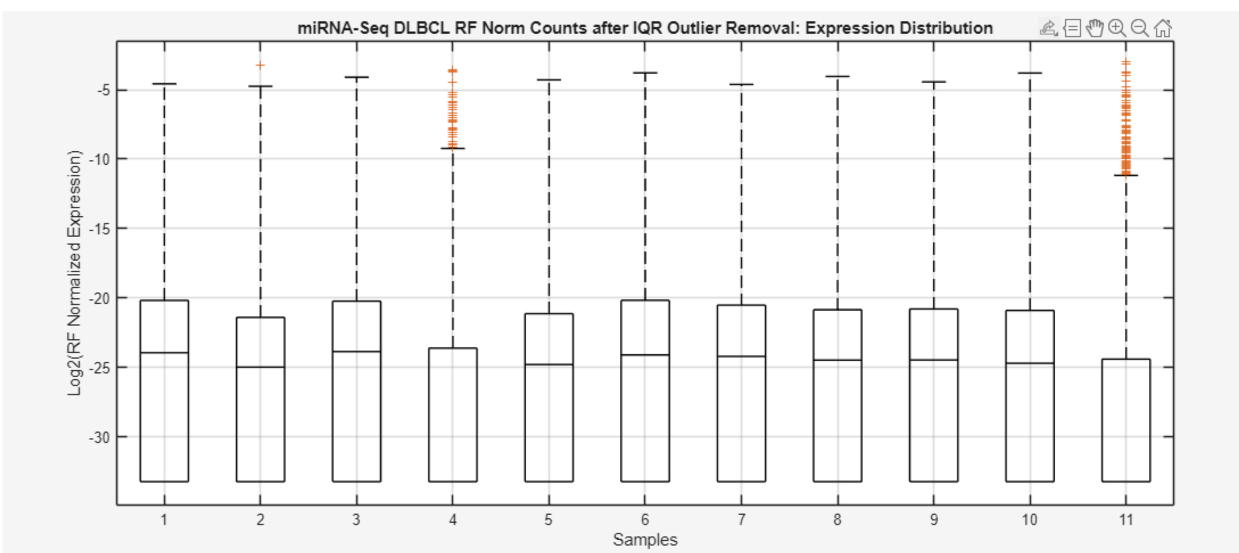

Figure 15. Boxplot of samples after removal of samples with IQR = 0.

Figure 13 demonstrated potential outliers in the dataset, with two samples having an IQR below the outlier threshold line. This was seen in sample 4 and sample 11. Figure 14 further supported this finding. After the removal of samples with IQR = 0, there was improved correlation between samples which now ranged from 0.7 to 0.8. However, sample 4 and 11 again were below the alpha threshold with a value nearing 0.45. Using course content, these samples were identified as outliers as they were 0.2 correlations away from the alpha threshold. The boxplots in Figure 15 further demonstrate differences of sample 4 and 11 from the rest of the samples, with the medians of all other samples being aligned. Moreover, there are numerous outliers above the upper whisker for samples 4 and 11. For these reasons, both samples 4 and 11 were removed from the dataset.

Data was visualized again after the removal of outliers, as seen in Figures 16, 17, and 18.

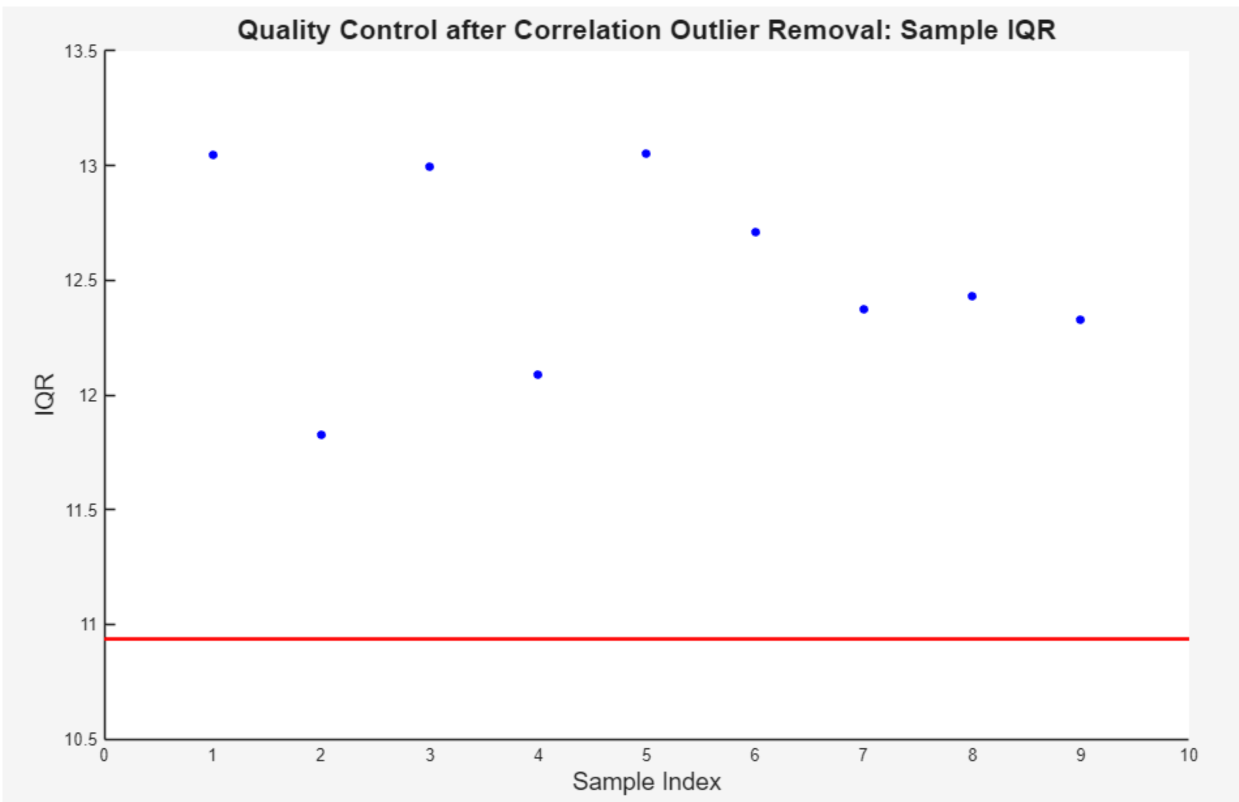

Figure 16. Scatterplot of IQR of samples after removal of outliers

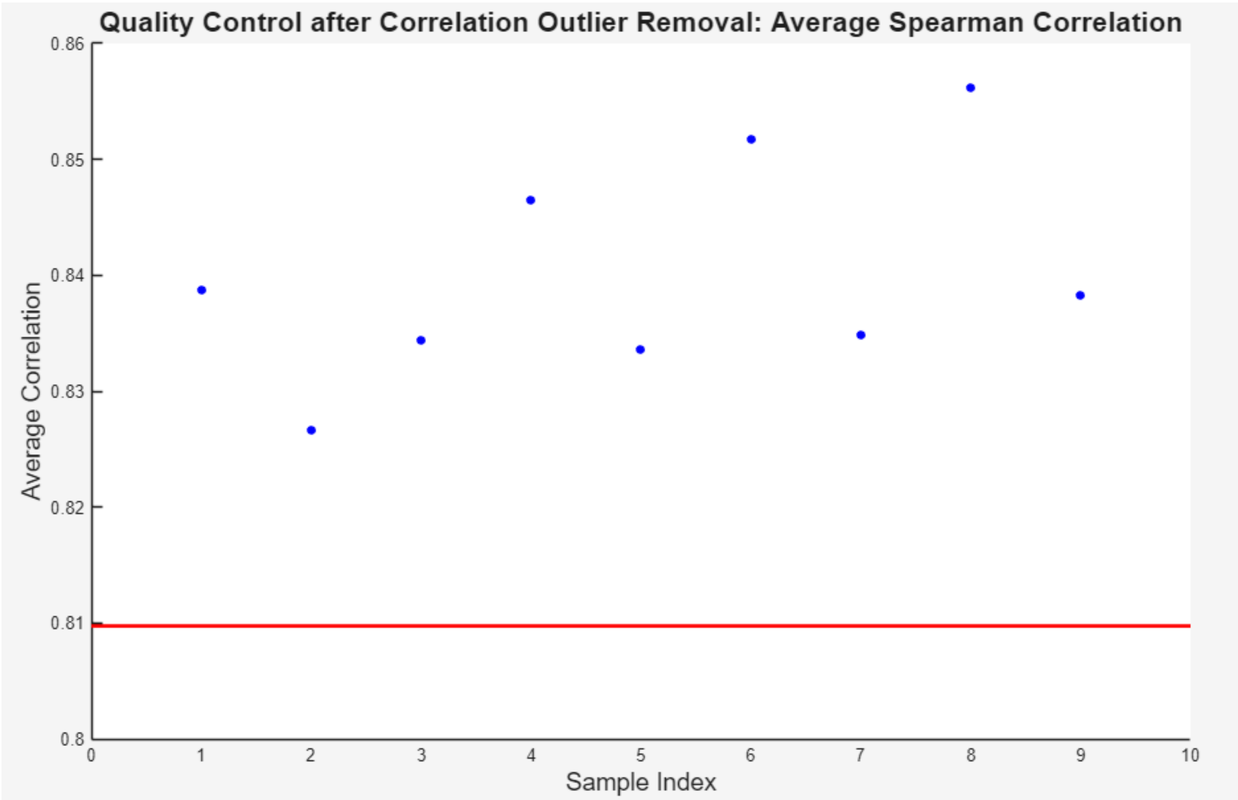

Figure 17. Scatterplot of correlation of samples after removal of outliers.

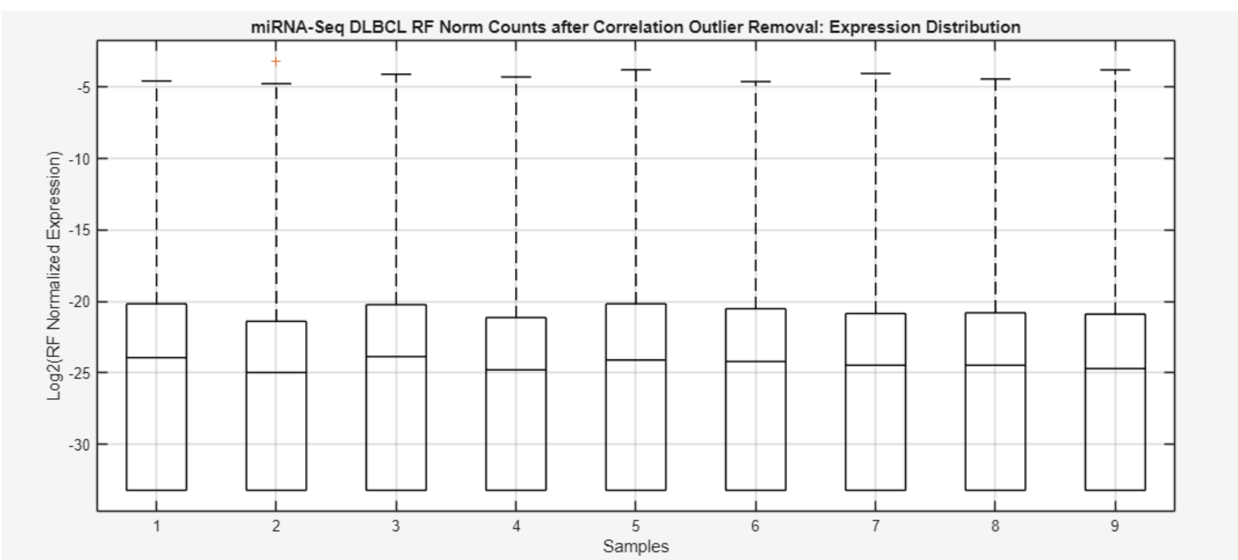

Figure 18. Boxplot of samples after removal of outliers.

In Figure 16, 17, 18, samples were filtered with a 0.90 quantile threshold of expression in hopes to exclude lowly expressed miRNAs. A 0.90 quantile threshold was selected to ensure that highly expressed miRNAs are adequately represented, without being too strict that it excludes important and biologically relevant miRNAs. In these figures, sample miRNA expression values were highly correlated to one another, ranging in value from 0.82 to 0.86. Moreover, the boxplots of the remaining samples were highly aligned in terms of IQR and median values. The 0.90 quantile threshold preserves miRNAs that are potentially informative in their localized expression and also removes features that are lowly-expressed. Subsequent distribution plots for final expression below show higher median values once the lowly-expressed genes were filtered out.

With this, Figure 19, 20, 21, and 22 below illustrate the final visualizations of the sample data. The IQR scatterplot and the subsequent boxplot illustrate that the IQRs of each sample are aligned and similar to one another. Moreover, all samples are highly correlated with a value ranging from 0.88 to 0.93. We see that there are no more outliers present nearing the threshold.

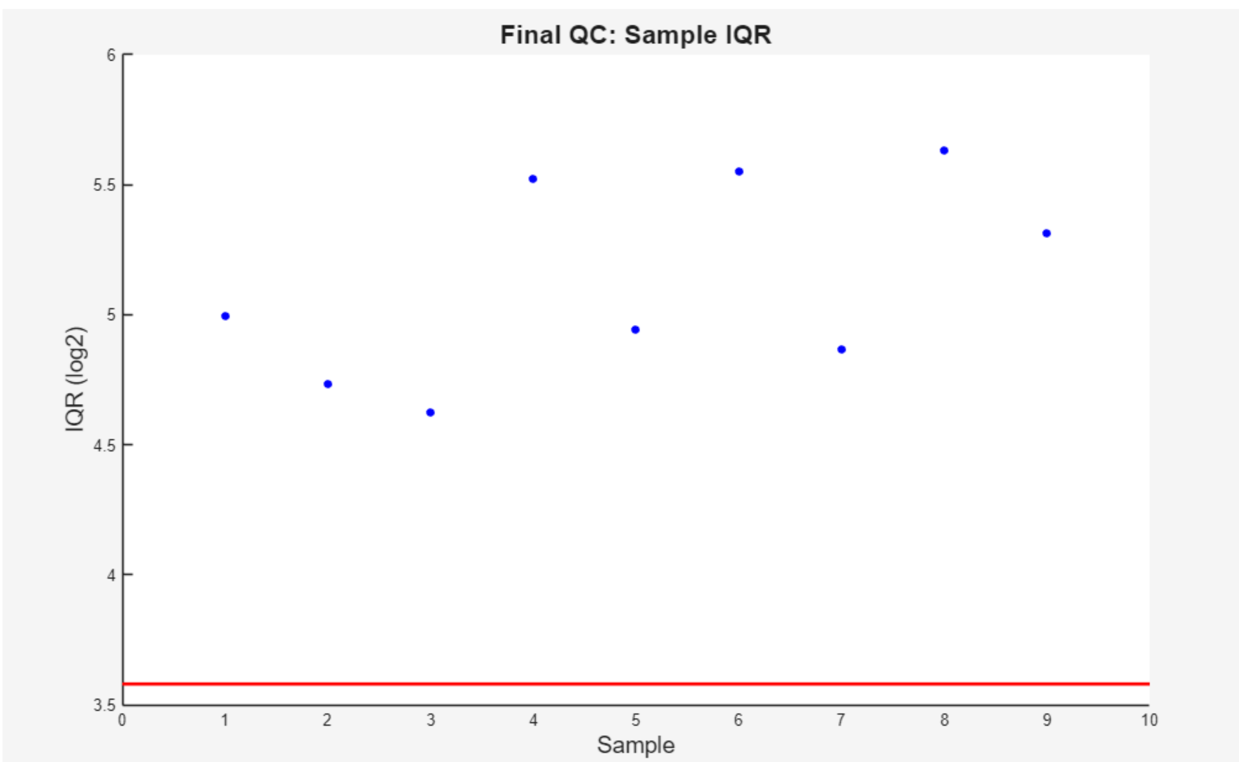

Figure 19. Final scatterplot of sample expression IQR.

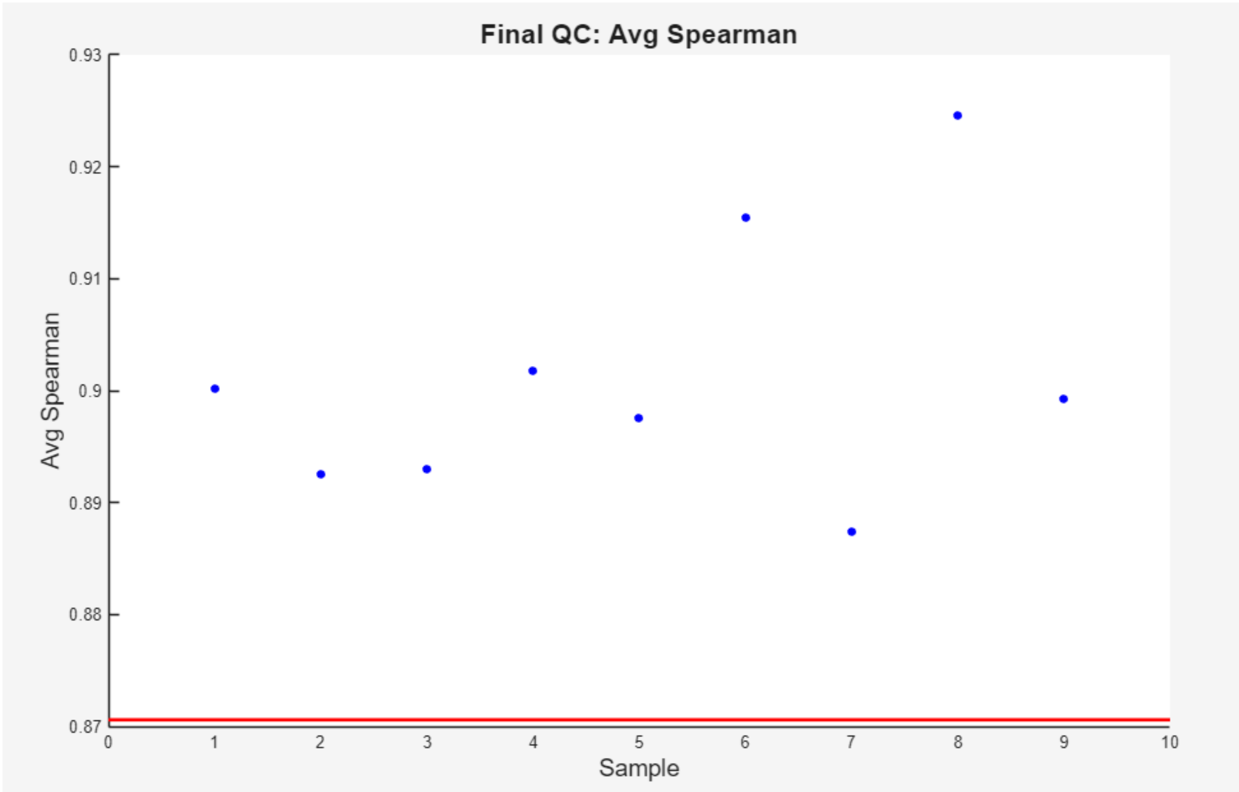

Figure 20. Final scatterplot of sample expression correlation.

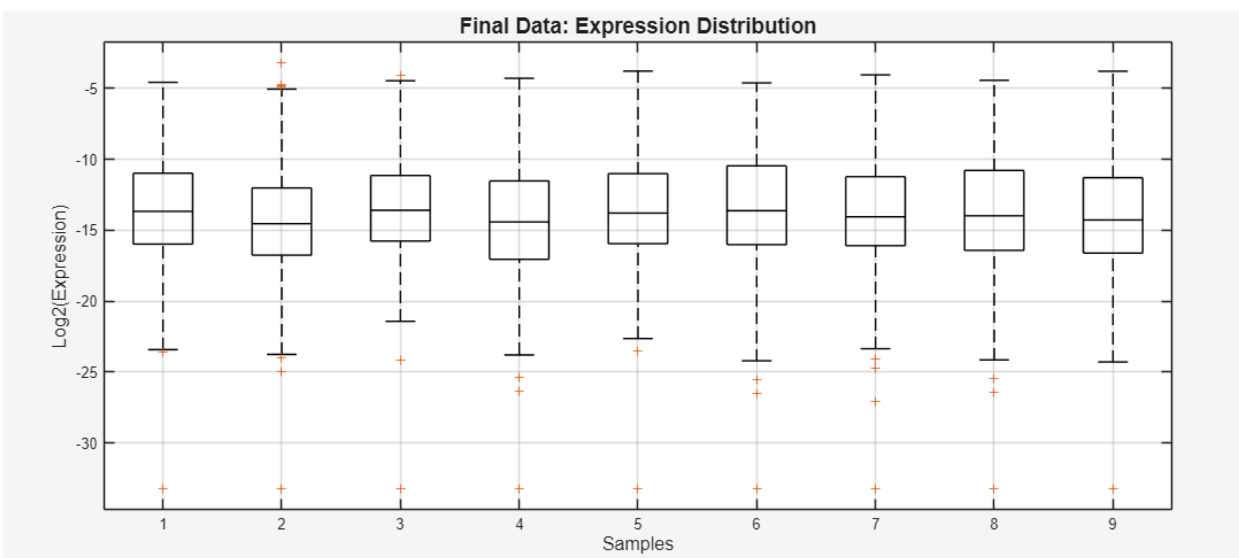

Figure 21. Final boxplot of sample expression.

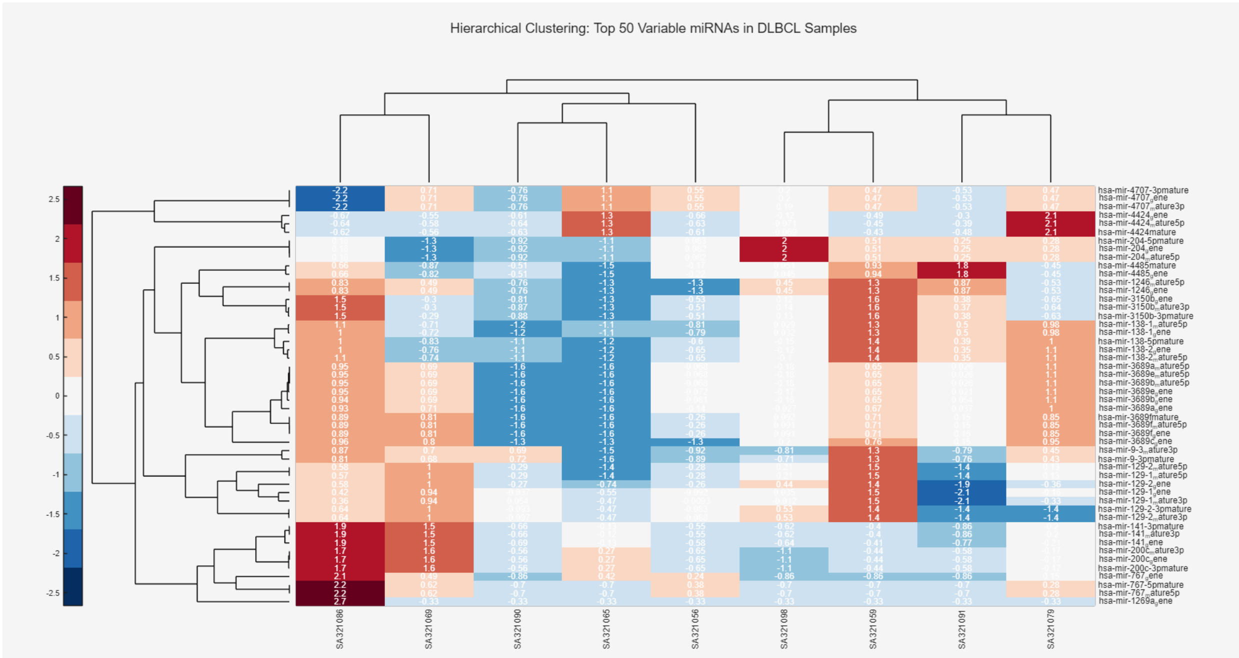

Figure 22. Hierarchical clustering of heatmap with filtered samples and top 50 miRNAs in DLBCL Samples.

In figure 22, the hierarchical clustering of the top 50 variable miRNAs to show the distinct grouping patterns in the filtered samples after the preprocessing and normalization steps. Each cell represents the relative expression level of individual miRNA (rows) across the samples (columns). We see colours which are warmer demonstrate a higher expression than that miRNA's median across its samples and we have a good grouping of clades indicating that there was not much noise picked up. There are also no batch stripes throughout the heatmap which suggests that the residual batch effects were minor and our QC looks successful thus far.

The clustering here is purely for visualization purposes, not for classification. Overall, final totals, IQR, and distribution plots have stable medians with consistent variability and no emerging outliers.

PART B

Batch	Region	Sample RNAseq ID
Batch 1	LEC	CUIS09 (AD-Braak) CUIS10 (AD-Braak) CUIS01 (Control) CUIS02 (Control) CUIS17 (PART)
	MEC	CUIS33 (AD-Braak) CUIS25 (Control) CUIS26 (Control) CUIS41 (PART)
	CB	CUIS57 (AD-Braak) CUIS49 (Control) CUIS50 (Control)
Batch 2	LEC	CUIS11 (AD-Braak) CUIS03 (Control) CUIS04 (Control)
	MEC	CUIS34 (AD-Braak) CUIS35 (AD-Braak) CUIS27 (Control) CUIS28 (Control) CUIS42 (PART)
	CB	CUIS58 (AD-Braak) CUIS51 (Control) CUIS52 (Control) CUIS65 (PART)
Batch 3	LEC	CUIS12 (AD-Braak) CUIS05 (Control) CUIS06 (Control) CUIS18 (PART)
	MEC	CUIS36 (AD-Braak) CUIS29 (Control) CUIS30 (Control)
	CB	CUIS59 (AD-Braak) CUIS60 (AD-Braak)

		CUIS53 (Control) CUIS54 (Control) CUIS66 (PART)
Batch 4	LEC	CUIS13 (AD-Braak) CUIS14 (AD-Braak) CUIS07 (Control) CUIS08 (Control) CUIS19 (PART)
	MEC	CUIS37 (AD-Braak) CUIS31 (Control) CUIS32 (Control) CUIS43 (PART)
	CB	CUIS61 (AD-Braak) CUIS55 (Control) CUIS56 (Control)
Batch 5	LEC	CUIS15 (AD-Braak) CUIS21 (Control) CUIS22 (Control)
	MEC	CUIS38 (AD-Braak) CUIS39 (AD-Braak) CUIS45 (Control) CUIS46 (Control) CUIS44 (PART)
	CB	CUIS62 (AD-Braak) CUIS69 (Control) CUIS70 (Control) CUIS67 (PART)
Batch 6	LEC	CUIS16 (AD-Braak) CUIS23 (Control) CUIS24 (Control) CUIS20 (PART)
	MEC	CUIS40 (AD-Braak) CUIS47 (Control) CUIS48 (Control)
	CB	CUIS63 (AD-Braak) CUIS64 (AD-Braak) CUIS71 (Control) CUIS72 (Control) CUIS68 (PART)

With an objective to compare gene expression across three categories: Control, AD-Braak, and Part, within 3 brain regions: LEC, MEC, and CB. With 72 samples in total, each batch can process 12 samples per run, with 6 batches overall. Since batch effects can mask biological difference, there has to be careful experimental design to balance brain regions across batches and diagnostic types. 2 controls were placed per region in LEC, MEC, CB for every batch so a baseline was available for every run, and AD-Braak and PART samples were rotated across each region and batch to minimize batch effects.

Each batch includes a proportional number of samples from all the groups, which prevents a single batch from constraining only a single tissue type or disease group, which might reduce batch effects. For each region and diagnosis layer there are random sample IDs that are randomly assigned to batches, ensuring that minor technical fluctuations are randomly distributed. Each batch has 4 samples per brain region, to keep tissue composition uniform across runs. There is no single diagnosis type that is processed continuously, instead categories are interlayered between runs in order to minimize systemic drift over time.

This design ensures that biological variability is preserved as all categories and regions appear equally across all batches. The randomization of batches reduces the operator specific bias, and replication across batches helps with reproducibility. By distributed samples in a balanced and randomized manner, the batch effects are minimized without having to sacrifice statistical tests.

References

Chapter II Neoplasms (C00-D48). ICD-10. https://icd.who.int/browse10/2019/en#C83.0